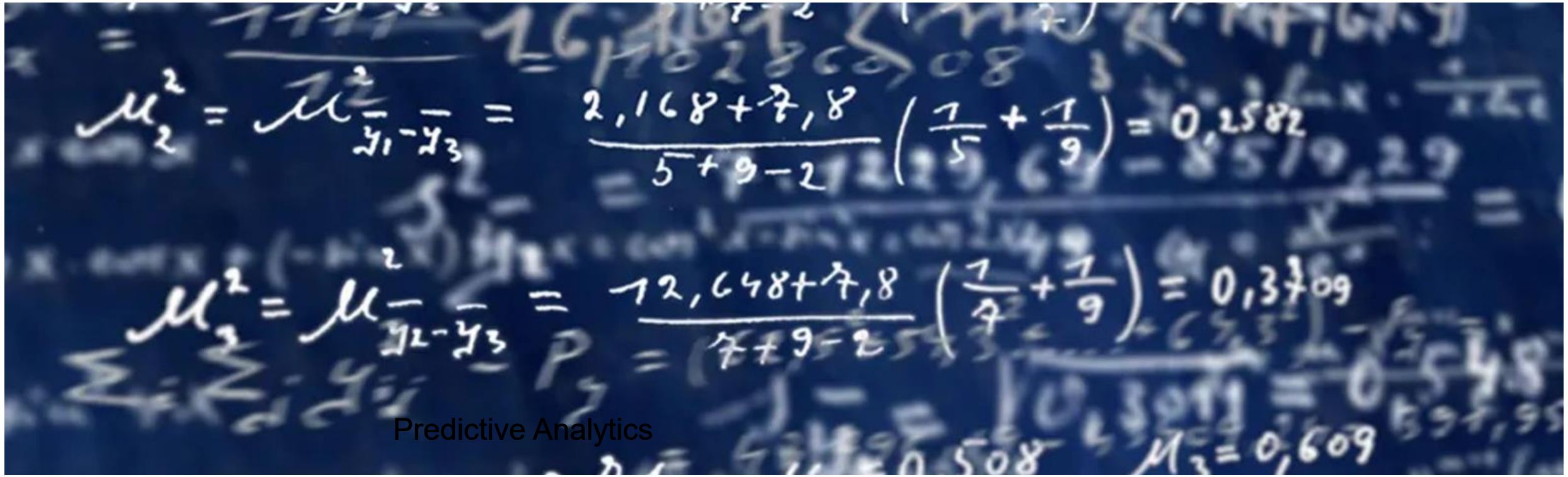


Predictive Analytics · Lecture 07

WORKING WITH ARIMA MODELS

CBS
COPENHAGEN BUSINESS SCHOOL
HANDELSHØJSKOLEN

F26

AGENDA

- ARIMA model basics
- Stationarity
- Testing for stationarity
- Model estimation
- Fooling around with a demo

ARIMA MODEL BASICS

THE BACKSHIFT OPERATOR B - AKA THE LAG OPERATOR L

- The backshift operator **B** shifts the time index backwards by one period:

$$By_t = y_{t-1}$$

- Conventional algebraic operations apply. Here are some useful examples:

$$B^2y_t = B \cdot By_t = By_{t-1} = y_{t-2}$$

$$B^n y_t = y_{t-n}$$

$$(1 - B)\, y_t = y_t - y_{t-1} = \Delta y_t$$

AN EXAMPLE OF THE USEFULNESS OF THE BACKSHIFT OPERATOR B – THE RANDOM WALK

The random walk, rewritten the tedious way:

$$\Delta y_t = \varepsilon_t \Leftrightarrow$$

$$y_t - y_{t-1} = \varepsilon_t \Leftrightarrow$$

$$y_t = \varepsilon_t + y_{t-1} = \varepsilon_t + \varepsilon_{t-1} + y_{t-2} = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} + y_{t-3} \Leftrightarrow$$

$$\boxed{y_t = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} + \varepsilon_{t-3} + \ldots}$$

The random walk, rewritten using the backshift operator:

$$\Delta y_t = \varepsilon_t \Leftrightarrow$$

$$(1 - B)y_t = \varepsilon_t \Leftrightarrow$$

$$y_t = \frac{1}{1 - B}\varepsilon_t = (1 + B + B^2 + B^3 + \ldots)\,\varepsilon_t$$

$$\boxed{y_t = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} + \varepsilon_{t-3} + \ldots}$$

THE BACKSHIFT OPERATOR ALLOWS FOR CONCISE REPRESENTATION OF ARIMA MODELS

The general ARIMA model:

$$\underbrace{(1 - \phi_1 B + \ldots + \phi_p B^p)}_{\text{AR-component of order } p}\ \underbrace{(1 - B)^d}_{\text{Integration component of order } d}\ x_t = \underbrace{(1 + \theta_1 B + \ldots + \theta_q B^q)}_{\text{MA-component of order } q}\ \varepsilon_t$$

Autoregressive component (AR) component

- The current value depends on its past values.
- Regression on lagged values of the variable itself.
- Order denoted by **p** - the number of time lags of the variable included in the model.

Integration (I) component

- Involves differencing the time series to achieve *stationarity*.
- Degree of differencing denoted by **d** - the number of times differencing is applied.
- Actual model variable is the **d'th** difference between consecutive values.

Moving Average component (MA)

- The current value of the model variable depends on current and past values of the error terms.
- Order denoted by **q** - the number of lagged errors included.

ARIMA EXAMPLES

ARIMA(0, 0, 0)
White Noise: $y_t = \varepsilon_t$

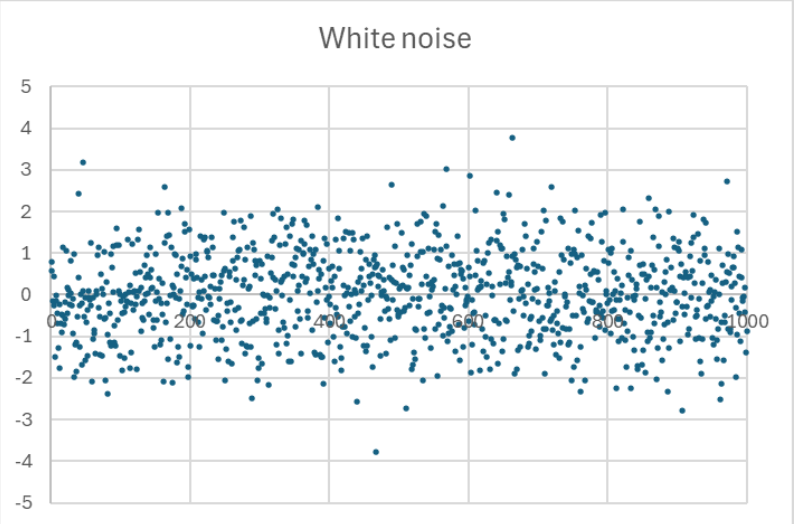
ARIMA(0, 1, 0)
Random Walk: $y_t = y_{t-1} + \varepsilon_t$

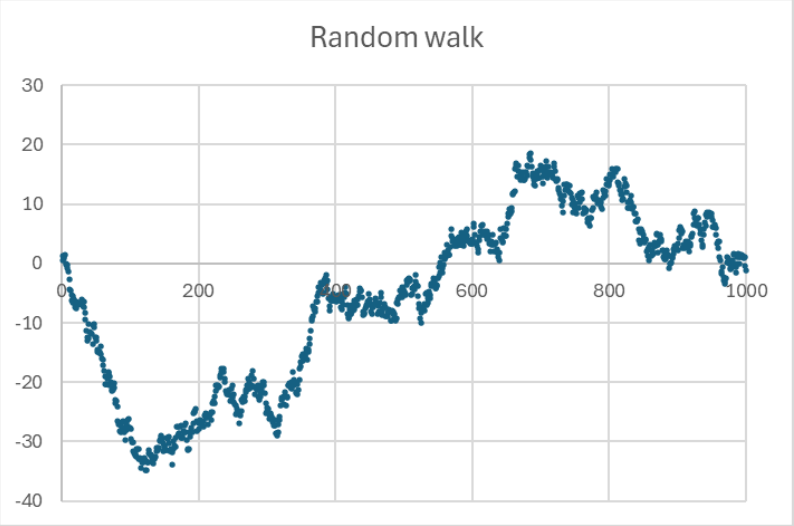
ARIMA(p, 0, 0)
Autoregressive AR(p): $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t$

ARIMA(0, 0, q)
Moving Average MA(q): $y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_q \varepsilon_{t-q}$

ARIMA(0, 1, 1)
Simple Exp. Smoothing:* $y_t = y_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t$ (with $\alpha = 1 - \theta_1$)

*) Simple Exponential Smoothing is just the optimal forecasting method for an ARIMA(0,1,1) process.



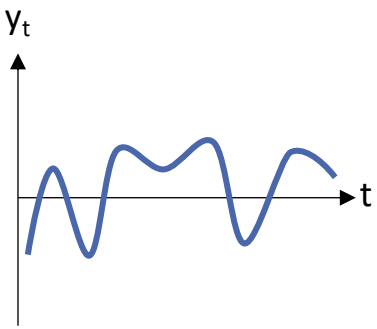


STATIONARITY

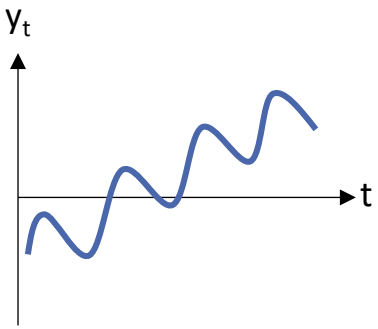
WHAT IS STATIONARITY

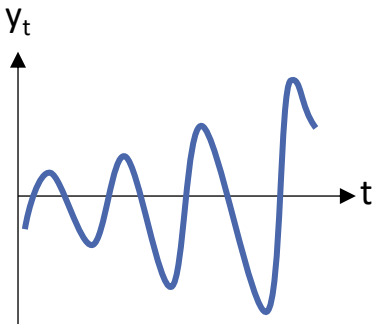
- Stationary series: Mean and variance are constant over time.
- Non-stationary series: Mean and/or variance changes over time (e.g. a random walk).

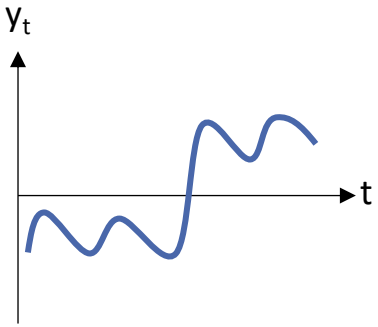
A stationary series

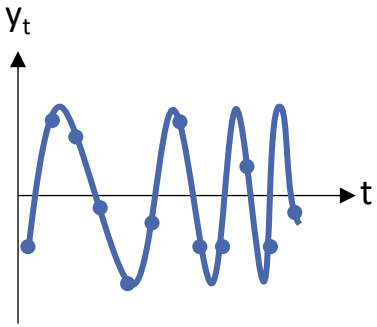


Nonstationary series









WHY IS STATIONARITY IMPORTANT?

- Most **ARIMA models** and other forecasting techniques require stationary data.
- So, before modeling, we must inspect the time series and test it to check whether we need to **transform** the data.
    - Log transformation
    - Differencing the time series

TYPES OF NON-STATIONARITY 1 - DETERMINISTIC TREND

- **Definition**

  The time series follows a **deterministic trend**, such as a straight line over time.

- **Model example**

$$y_t = \alpha + \beta t + \varepsilon_t$$

  where ε_t is a stationary error term.

- **Detrending**

  Remove the trend by regression or differencing once (if linear) or several times (if exponential).

A time series with a deterministic trend is said to be **trend-stationary** if, after removing the trend, the residuals are stationary

TYPES OF NON-STATIONARITY 2 - STOCHASTIC TREND (UNIT ROOT)

- **Definition**

  The time series **accumulates shocks** over time, so the mean **drifts** unpredictably — the shocks accumulate permanently.

- **Model example**

$$y_t = y_{t-1} + \varepsilon_t \quad \text{or} \quad (1 - B)\, y_t = \varepsilon_t$$

  where ε_t is a stationary error term.

- **Detrending**

  Differencing.

A time series with a stochastic trend is **not trend-stationary,** because there is no particular trend to remove.

TYPES OF NON-STATIONARITY 3 - DRIFT

- **Definition**

  The series has a **constant additive component** (not necessarily a linear time trend).

- **Model example**

$$y_t = \mu + y_{t-1} + \varepsilon_t$$

  where ε_t is a stationary error term. The example is a **unit root process with drift**. It's still **non-stationary**

- **Solution**

  First differencing removes both the drift and unit root.

TYPES OF NON-STATIONARITY 4 - STRUCTURAL BREAKS (REGIME SHIFTS)

- **Definition**: The statistical properties change **suddenly** at some point in time (e.g. after a policy change or an economic crisis).

- Breaks can appear in mean, variance or trend.

- **Solution**
  Model with **dummy variables** for breakpoints…
  … or **segment** the series.

TYPES OF NON-STATIONARITY 5 - HETEROSKEDASTICITY

- **Definition**

  The **variance** of the series **changes over time**.

  - Common in lots of economic and financial series that are subject to growth and/or inflation.
  - In some series – particularly in finance – the variance (or volatility) 'clusters'. Such series could be modelled according to a GARCH process.

- **Solution**

  Variance-stabilizing transforms, **e.g. log transformation…**

  …or model the variance directly, **e.g. with GARCH.**

GARCH processes are beyond the scope of this course

SUMMARY OF NON-STATIONARITY

TYPE OF NON-STATIONARITY	CAUSE	FIX / TRANSFORMATION
Trend non-stationarity	Deterministic trend (e.g., linear)	Detrending or differencing
Difference non-stationarity (unit root)	Stochastic trend / accumulated shocks	Differencing
Drift	Constant additive trend	Differencing
Structural breaks	Sudden changes in structure	Breakpoint modeling or dummies
Variance non-stationarity (heterosked.)	Changing volatility over time	Log transformation / GARCH models

TESTING FOR STATIONARITY

STATISTICAL TESTING

*A statistical test evaluates whether **observed sample data** are consistent with **a hypothesized value of a parameter of the datagenerating process.***

1. Hypotheses

- Null hypothesis, H_0 – A baseline claim about a DGP parameter, e.g. $\mathbf{H_0}: \mu = \mu_0$
- Alternative hypothesis, H_1 – A competing claim, e.g. $\mathbf{H_1}: \mu \neq \mu_0$ or $\mathbf{H_1}: \mu < \mu_0$ or $\mathbf{H_1}: \mu > \mu_0$

2. Test statistic

- A statistic computed **from the sample** that measures the distance between data and H_0, e.g.

$$\mathbf{z} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

*Under H_0, **z** has a known probability distribution*

3. Two equivalent decision rules

Critical value approach

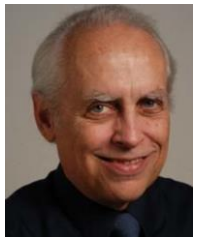
- Choose **significance level (test size)** α
- Determine **critical value** from the sampling distribution
- **Reject H_0** if the test statistic falls in the **rejection region**

p-value approach

- Compute **p-value**, the probability of observing a statistic at least as extreme as the one observed **if H_0 were true**
- **Reject H_0** if p-value < α

THE AUGMENTED DICKEY-FULLER (ADF) TEST

David Dickey



Wayne A. Fuller

*Testing for a **unit root***
*— in other words, testing whether the series is **non-stationary**.*

What does the test actually do?

H_0: The series has a unit root (i.e. is non-stationary).

H_1: The series is stationary.

Test model

Estimate $$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i y_{t-1} + \varepsilon_t$$

- p *lags included to "augment" the test*
- *The key coefficient —* H_0: $\gamma = 0$*, indicating unit root vs.* H_1: $\gamma < 0$*, no unit root*
- *Time trend (optional)*
- *Constant term (optional)*
- *First difference of series*

PLEASE NOTE:

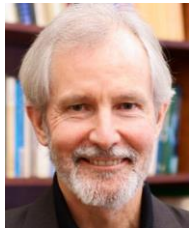
The ADF test does not report a regular *t-statistic*. It uses **Dickey-Fuller critical values**, which are **not** the same as for a regular t-test — so you need:

- rely on the **ADF test's p-value**, or
- compare the test statistic to the special **critical values** (often shown in the output).

THE KPSS TEST

Andrzej Kwiatkowski

Peter C.B. Phillips



Peter Schmidt

Yongcheol Shin

*Testing for **stationarity***
*— the **opposite** of the Augmented Dickey-Fuller (ADF) test.*

What does the test actually do?

H_0: The series is **stationary** (around a mean or around a deterministic trend).

H_1: The series is **non-stationary** (i.e., has a unit root).

Two versions of the test

1. Level Stationarity Test: Tests whether the series is stationary around a constant (mean).
2. Trend Stationarity Test: Tests whether the series is stationary around a deterministic trend.

In R, you can select this using the type argument: "mu" (level) or "tau" (trend).

Test model and interpretation

The KPSS model decomposes the series as $y_t = r_t + \varepsilon_t$, where r_t is a **random walk** (possibly zero under the null).
If r_t is constant (i.e. no random walk), the series is **stationary.**

- If the **KPSS test statistic is greater** than the **critical value**, we **reject the null** — the series is **non-stationary**.
- If the **test statistic is small**, we **fail to reject the null** — the series is **stationary**.

PLEASE NOTE:

Like the ADF test, the KPSS test statistic is not normally distributed. You compare it to **critical values** provided by KPSS tables or automatically reported in software.

THE AUTOCORRELATION FUNCTION

*The **Autocorrelation Function (ACF)** measures the*
correlation between a time series and lagged versions of itself:

- $ACF(k) = Corr(y_t, y_{t-k})$
- ACF at lag k shows how strongly the series is correlated with its past k steps.

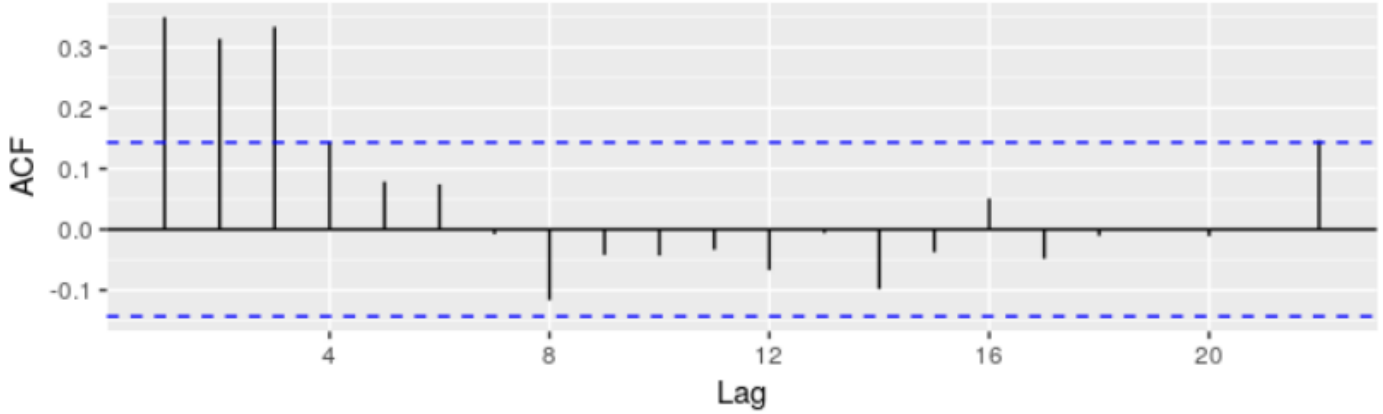


Figure 8.9: ACF of quarterly percentage change in US consumption.

THREE USES OF THE ACF

1. Identify Stationarity

- A ***stationary*** series typically has rapidly decaying ACF (small autocorrelations after a few lags).
- A ***non-stationary*** series shows slowly decaying or persistently high ACF (especially at low lags).

2. Determine Order of MA(q) Processes

- For a **Moving Average** process of order q (MA(q)), the ACF cuts off after lag q, *i.e.* ***ACF becomes appr. zero beyond lag q***.
- This helps determine how many lags to include in the MA part of an ARIMA model.

3. Check Residuals of a Fitted Model

- Plot the ACF of the residuals:
  - If the residuals are white noise (i.e., uncorrelated), the ACF should show no significant spikes.
  - Significant autocorrelation in residuals → model might be misspecified.

THE PACF (PARTIAL AUTOCORRELATION FUNCTION)

*The **Partial Autocorrelation Function (PACF)** measures the **correlation between a time series and its lagged values, after removing the effects of shorter lags**.*

- The PACF shows the **direct effect** of y_{t-k} on y_t, after controlling for lags 1 through k−1.
- This helps isolate the unique contribution of each lag.

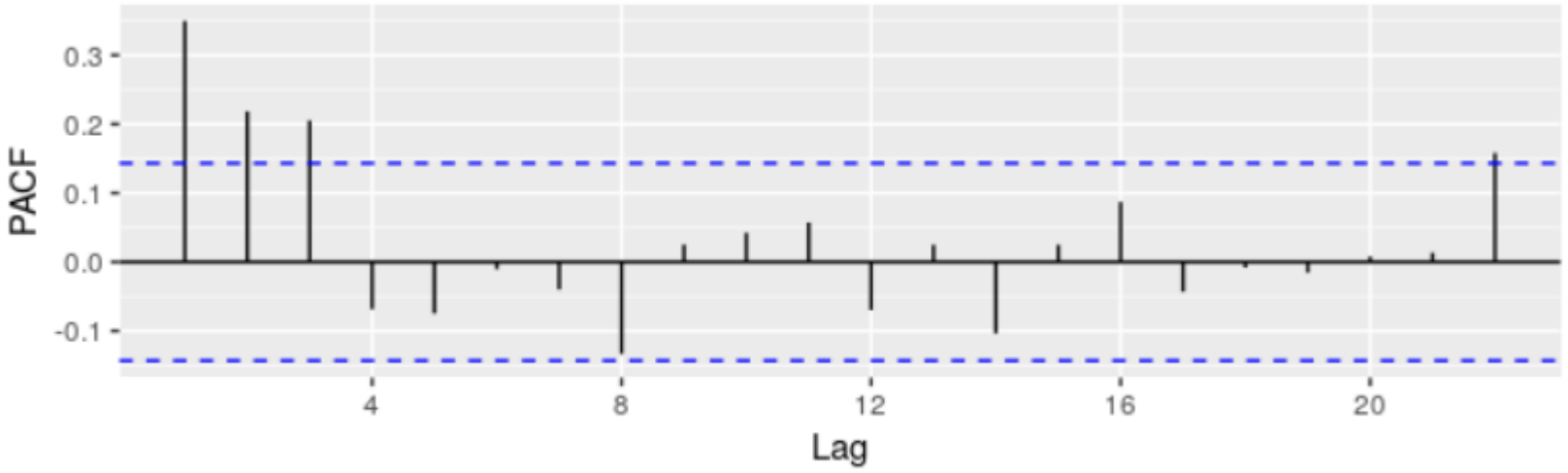


Figure 8.10: PACF of quarterly percentage change in US consumption.

HOW TO USE THE PACF

1. Identify Stationarity

- In a *stationary series*, PACF values typically drop off quickly (i.e., no persistent significant lags).
- In a *non-stationary* series, PACF can show strong autocorrelation at **low lags**.

2. Determine Order of AR(p) Processes

- For an **Autoregressive** process of order p (AR(p)), the PACF cuts off after lag p, *i.e.* ***PACF becomes appr. zero beyond lag p***.
- Meaning: significant values up to lag p, and approx. zero beyond.
- This helps determine how many lags to include in the AR part of an ARIMA model.

3. Model Diagnostics (Residual Checking)

- Plot the ACF of the residuals:
  - If the residuals are white noise, no significant spikes should appear in the PACF.
  - Significant PACF values suggest **that the model is missing AR components**.

SUMMARY

Model Type	ACF Pattern	PACF Pattern
AR(p)	Tails off	Cuts off at lag p
MA(q)	Cuts off at lag q	Tails off
ARMA(p,q)	Both tail off	Both tail off

…and if the ACF and PACF is not tailing off, the model is probably not stationary!

MODEL ESTIMATION

OBJECTIVE & RECIPE

Objective

- To estimate the p + q parameters of the ARIMA-model…
- In a **parsimonious** way, i.e. with as few parameters as possible.

Recipe

1. Inspect the timeseries for stationarity.
2. Apply any transformations necessary to obtain a stationary series.
3. Test for stationarity.
   If still not stationary, go to #2.
4. Analyze ACF and PACF to get a grip on orders of AR and MA components, i.e. p and q.
5. Estimate model.
6. Check goodness of fit, including residuals.
   If serial correlation remains in error term, go to #4

GOODNESS OF FIT-MEASURES

AIC (Akaike Information Criterion)

- Measures **model fit** and **complexity**.
- Lower AIC = **better** model (among the options).

BIC (Bayesian Information Criterion)

- Similar to AIC, but **penalizes model complexity more heavily**.
- Lower BIC = **better** if you want to avoid overfitting.

Model selection rule of thumb:

- If **both AIC and BIC are lower for the same model**, it's clearly preferred.
- If AIC prefers one model and BIC prefers another, you decide based on whether you're prioritizing:
    - **Better fit** (choose AIC),
    - or **Simpler model** (choose BIC).

FOOLING AROUND IN R-STUDIO

Thomas Einfeldt, Associate Teaching Professor
Økonomisk Institut
Copenhagen Business School

Porcelænshaven 16A, 1. sal
2000 Frederiksberg
tce.eco@cbs.dk